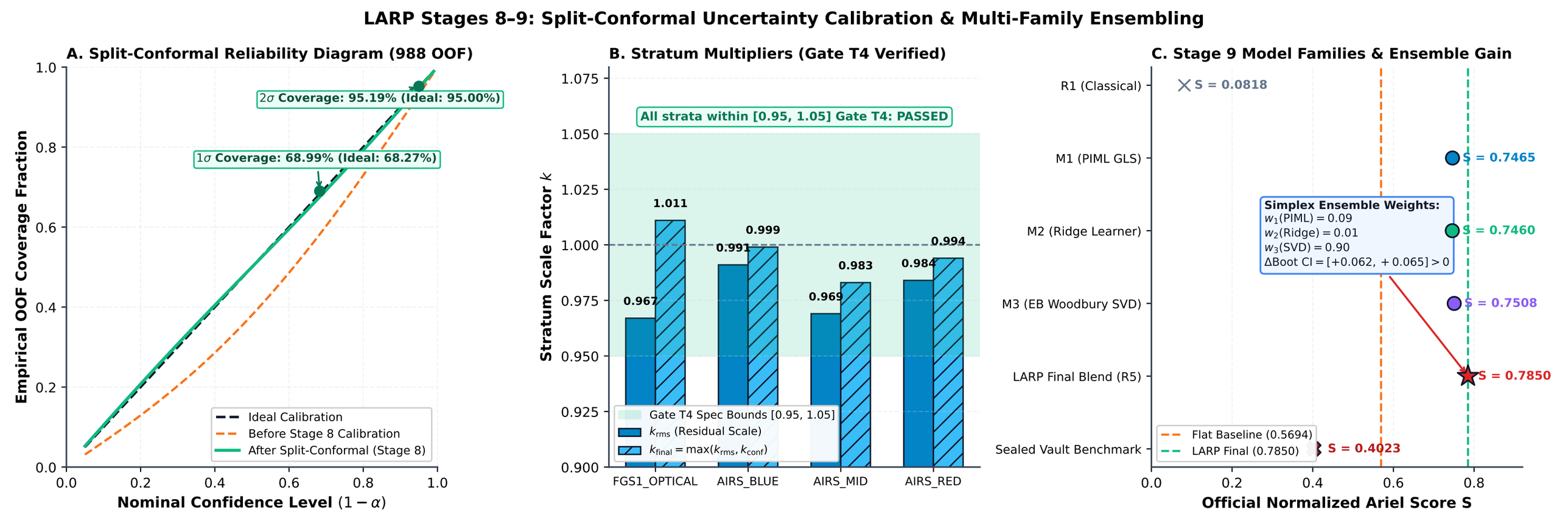
LARP Stages 8–9: Split-Conformal Uncertainty Calibration & Multi-Family Ensembling
A. Split-Conformal Reliability Diagram (988 OOF)
2σ Coverage: 95.19% (Ideal: 95.00%)
1σ Coverage: 68.99% (Ideal: 68.27%)
Ideal Calibration
Before Stage 8 Calibration
After Split-Conformal (Stage 8)
Empirical OOF Coverage Fraction
Nominal Confidence Level (1 − α)
B. Stratum Multipliers (Gate T4 Verified)
All strata within [0.95, 1.05] Gate T4: PASSED
Gate T4 Spec Bounds [0.95, 1.05]
k_rms (Residual Scale)
k_final = max(k_rms, k_conf)
Stratum Scale Factor k
1.011
0.967
0.999
0.991
0.983
0.969
0.994
0.984
FGS1_OPTICAL
AIRS_BLUE
AIRS_MID
AIRS_RED
C. Stage 9 Model Families & Ensemble Gain
R1 (Classical)
S = 0.0818
M1 (PIML GLS)
S = 0.7465
M2 (Ridge Learner)
S = 0.7460
M3 (EB Woodbury SVD)
S = 0.7508
LARP Final Blend (R5)
S = 0.7850
Sealed Vault Benchmark
S = 0.4023
Simplex Ensemble Weights:
w1(PIML) = 0.09
w2(Ridge) = 0.01
w3(SVD) = 0.90
ΔBoot CI = [+0.062, +0.065] > 0
Flat Baseline (0.5694)
LARP Final (0.7850)
Official Normalized Ariel Score S